

COMPANION SLIDES

Chapter 11

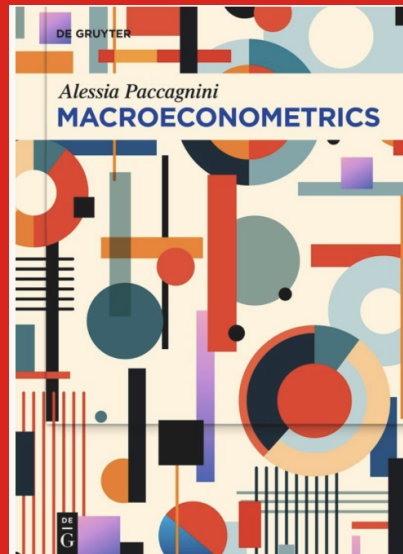
Mixed-Frequency Data

Using information when it actually arrives

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Macroeconomic information arrives at different speeds

How can we combine monthly, quarterly, weekly and daily information without discarding useful timing information?

- ▶ GDP is quarterly, while many indicators arrive monthly or faster.
- ▶ Temporal aggregation can hide within-period dynamics.
- ▶ Mixed-frequency methods preserve information at its native frequency.

Frequency is part of the information set.

Core methods

- ▶ Mixed-frequency landscape
- ▶ Temporal aggregation bias
- ▶ MIDAS regressions
- ▶ Weighting schemes
- ▶ U-MIDAS
- ▶ MIDAS in VARs

Systems and applications

- ▶ Mixed-frequency VARs
- ▶ Bayesian shrinkage
- ▶ State-space formulations
- ▶ Structural analysis
- ▶ Nowcasting and ragged edges
- ▶ US GDP–FFR application

- ▶ Explain why temporal aggregation can destroy economically relevant information.
- ▶ Specify a MIDAS regression and interpret its weighting function.
- ▶ Distinguish parametric MIDAS from U-MIDAS.
- ▶ Construct the stacked representation of an MF-VAR.
- ▶ Explain how mixed-frequency models support structural analysis and nowcasting.
- ▶ Discuss ragged edges, publication lags and news revisions.
- ▶ Interpret the empirical comparison between MF-VAR and low-frequency VAR results.

Quarterly

GDP, national accounts

Monthly

employment, inflation,
production, policy rates

Weekly / Daily

financial prices, claims,
payments, mobility

Challenge

A quarterly model often forces faster variables to be averaged or sampled, even though their timing may contain predictive and structural information.

- ▶ Averaging high-frequency observations can smooth away turning points.
- ▶ End-of-period sampling can ignore information from earlier observations.
- ▶ Aggregation can alter persistence and cross-variable correlations.
- ▶ In structural analysis, it can distort the timing and apparent size of shocks.

Aggregation is not always an innocuous data-cleaning step.

$$y_{\tau} = \alpha + \beta \sum_{k=1}^K w(k; \theta) x_{\tau-k/m}^{(m)} + \varepsilon_{\tau}$$

- ▶ y_{τ} : low-frequency target.
- ▶ $x^{(m)}$: high-frequency predictor with m observations per low-frequency period.
- ▶ $w(k; \theta)$: parsimonious lag-weight function.
- ▶ MIDAS uses many high-frequency lags without estimating one unrestricted coefficient for every lag.

Without restrictions

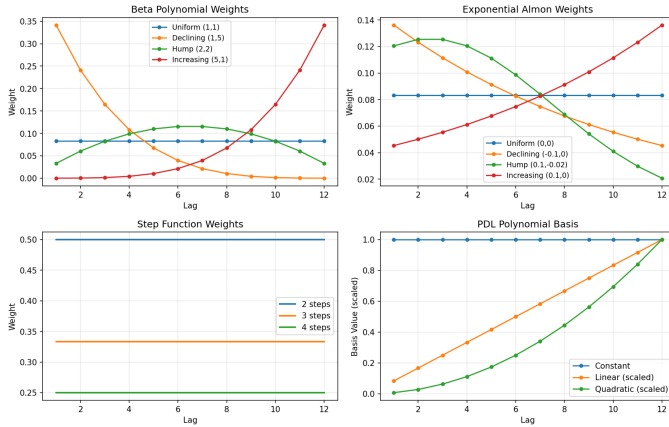
Many lags quickly create parameter proliferation and unstable estimates.

With MIDAS weights

A small parameter vector determines the shape of a long distributed lag.

Parsimony trades some flexibility for lower estimation variance.

Comparison of MIDAS Weighting Schemes



Parametric MIDAS

- ▶ smooth lag weights
- ▶ few parameters
- ▶ strong regularization
- ▶ useful with many lags

U-MIDAS

- ▶ one coefficient per lag
- ▶ more flexible
- ▶ parameter intensive
- ▶ greater overfitting risk

- ▶ Mixed-frequency information can also be embedded in a state-space system.
- ▶ The Kalman filter naturally handles missing observations and irregular availability.
- ▶ This is especially useful when latent common factors link many indicators observed at different frequencies.
- ▶ State-space approaches are central to operational nowcasting systems.

- ▶ **Nonlinear MIDAS**: allow asymmetric or state-dependent effects.
- ▶ **Density forecasting**: model predictive distributions, not only conditional means.
- ▶ **Finance**: combine daily/weekly financial predictors with monthly or quarterly outcomes.
- ▶ **Macroeconomics**: exploit timely indicators for forecasting and real-time monitoring.
- ▶ **MIDAS-VAR**: bring mixed-frequency information into multivariate systems.

$$Z_{\tau} = \begin{pmatrix} x_{\tau,1} \\ x_{\tau,2} \\ x_{\tau,3} \\ y_{\tau} \end{pmatrix}, \quad Z_{\tau} = c + A_1 Z_{\tau-1} + \cdots + A_p Z_{\tau-p} + u_{\tau}$$

- ▶ High-frequency observations are stacked within each low-frequency period.
- ▶ The system becomes a standard VAR in an expanded vector.
- ▶ Standard VAR tools can then be used directly.

- ▶ Stacking preserves timing, but increases the number of variables.
- ▶ Multiple high-frequency indicators can make the VAR very large.
- ▶ More variables and lags imply many coefficients and covariance parameters.
- ▶ **Bayesian shrinkage** can stabilize large MF-VARs by shrinking weak coefficients toward structured priors.

Stacked MF-VAR

Observation-driven; transparent; standard VAR estimation and identification apply.

State-space formulation

Flexible with missing data and ragged edges; naturally uses Kalman filtering.

The right representation depends on the question and data-release structure.

- ▶ Structural shocks can be identified in the expanded mixed-frequency system.
- ▶ IRFs can reveal whether the timing of a shock within a quarter matters.
- ▶ Temporal alignment is crucial: month 1, month 2 and month 3 shocks are not necessarily equivalent.
- ▶ Aggregating before identification can conflate predictable within-quarter dynamics with genuine surprises.

The nowcasting problem

Estimate current-quarter activity before the official low-frequency release arrives.

- ▶ Monthly and weekly indicators arrive throughout the quarter.
- ▶ A nowcast should update as each new release becomes available.
- ▶ Mixed-frequency models exploit these releases without waiting for a balanced quarterly dataset.

- ▶ At a given date, different series end at different observations.
- ▶ Some monthly releases are current; others lag by one or more months.
- ▶ Daily and weekly financial series may be nearly current while GDP remains available only quarterly.
- ▶ State-space methods can treat these irregular endings as missing observations rather than discarding recent data.

A useful decomposition

A new release changes the nowcast when the realization differs from what the model expected.

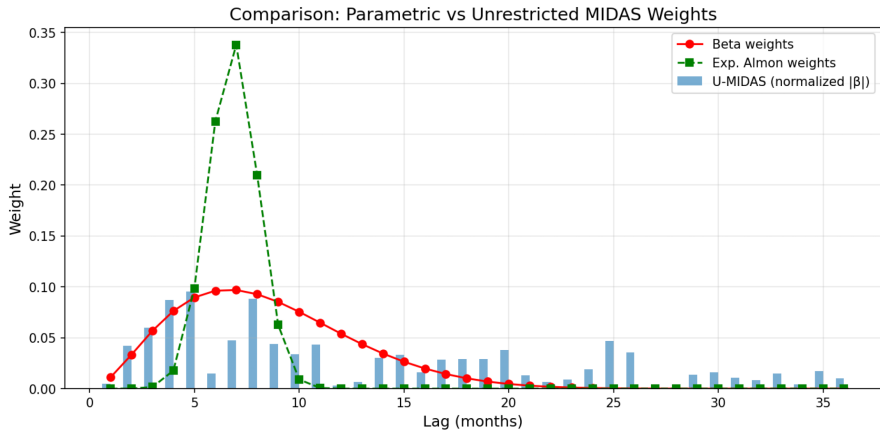
- ▶ **News**: release minus its forecast.
- ▶ **Weight**: predictive relevance of that release for the target.
- ▶ The product of news and weight explains the contribution to the nowcast revision.
- ▶ This decomposition helps communicate why a real-time assessment moved.

- ▶ Uncertainty and business-cycle analysis.
- ▶ Real-time macroeconomic monitoring.
- ▶ Output-gap nowcasting using monthly indicators inside an MF-BVAR.
- ▶ COVID-era monitoring with weekly/daily alternative data.
- ▶ Monetary policy shocks identified at high frequency but propagated to lower-frequency macro variables.

- ▶ Monthly effective Federal Funds Rate (FEDFUNDS): 1954:M7–2025:M12.
- ▶ Quarterly real GDP (GDPC1): 1947:Q1–2025:Q3.
- ▶ GDP growth is the annualized quarterly log difference.
- ▶ Estimation samples span roughly 1957:Q3–2025:Q3, depending on lag structure.
- ▶ The exercise compares MIDAS regressions with a stacked MF-VAR.

$$y_{\tau} = \alpha + \beta \sum_{k=1}^{12} w(k; \theta) x_{\tau-k/3}^{(3)} + \varepsilon_{\tau}$$

- ▶ $K = 12$ monthly FFR lags: four quarters of history.
- ▶ Parametric alternatives: Beta polynomial and exponential Almon weights.
- ▶ Benchmark: unrestricted MIDAS with separate lag coefficients.



Parametric MIDAS

- ▶ 4 parameters
- ▶ low R^2
- ▶ BIC about 818–820
- ▶ smooth lag decay

U-MIDAS

- ▶ 37 parameters
- ▶ $R^2 = 0.299$
- ▶ BIC = 910.48
- ▶ much greater flexibility

Higher in-sample fit does not automatically imply a better forecasting specification.

$$Z_{\tau} = (FFR_{m1,\tau}, FFR_{m2,\tau}, FFR_{m3,\tau}, GDP_{\tau})'$$

- ▶ The three monthly policy-rate observations remain separate.
- ▶ The chapter proceeds with a parsimonious MF-VAR(1) after lag-selection comparisons.
- ▶ The expanded system reveals within-quarter dynamics suppressed by quarterly averaging.

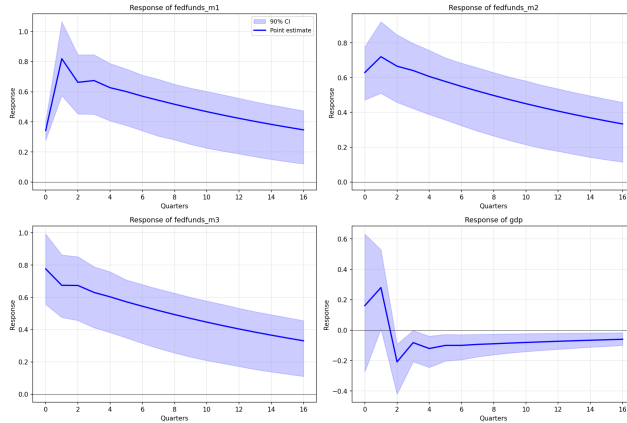
- ▶ Strong end-of-quarter FFR persistence: month 3 is highly informative for the next quarter's policy stance.
- ▶ Lagged GDP growth enters positively in the month-2 and month-3 FFR equations, consistent with policy feedback to economic conditions.
- ▶ Granger-causality tests reject non-causality in both directions at the 1% level.
- ▶ Mixed-frequency structure exposes feedback that quarterly aggregation can conceal.

Cholesky ordering

$[FFR_{m1}, FFR_{m2}, FFR_{m3}, GDP]$

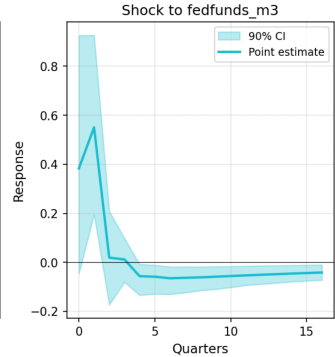
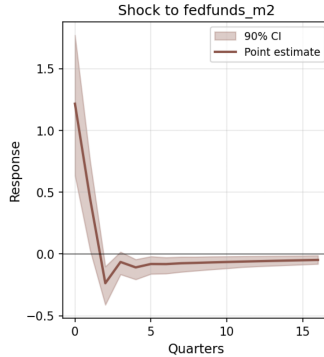
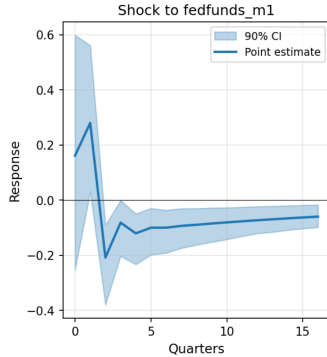
- ▶ Within-quarter interest-rate shocks may affect GDP contemporaneously.
- ▶ GDP shocks do not affect same-quarter interest rates under this ordering.
- ▶ The expanded system allows the timing of policy shocks within the quarter to be studied explicitly.

Impulse Responses to fedfunds_m1 Shock (1 Std. Dev.)
90% Confidence Intervals (Bootstrap)



- ▶ The impact response is weakly positive and statistically insignificant.
- ▶ The contractionary effect appears with a lag.
- ▶ At $h = 2$, the GDP response is about -0.21 percentage points and the 90% bootstrap interval excludes zero.
- ▶ The response remains negative through longer horizons, consistent with delayed monetary transmission.

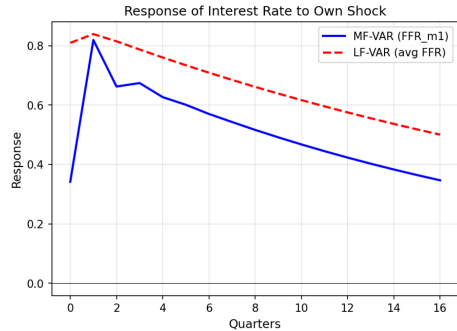
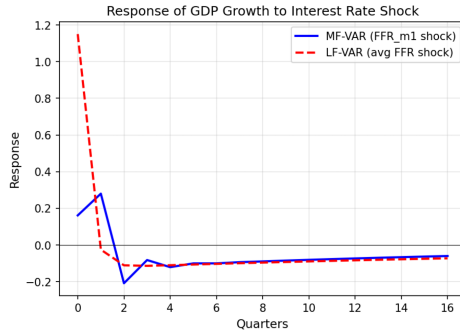
Response of gdp to Different Shocks 90% Confidence Intervals (Bootstrap)



- ▶ Month-2 FFR shocks have the largest contemporaneous GDP impact in the chapter application.
- ▶ Month-3 shocks display a more delayed contractionary profile.
- ▶ Short-run transmission therefore depends on when within the quarter the policy shock occurs.
- ▶ By longer horizons, the responses become more similar.

Quarterly averaging would hide this short-run heterogeneity.

MF-VAR vs LF-VAR: Impulse Response Comparison
(Illustrating potential temporal aggregation bias)



- ▶ The quarterly VAR produces a much larger immediate GDP response than the MF-VAR.
- ▶ It also attributes greater persistence to the interest-rate process.
- ▶ Averaging monthly rates mixes genuine policy surprises with predictable within-quarter movements.
- ▶ Structural conclusions can therefore depend on whether native-frequency information is preserved.

Choose MIDAS when...

the main goal is forecasting a low-frequency target using high-frequency predictors with parsimonious lag weights.

Choose MF-VAR when...

the question involves feedback, Granger causality, shock identification, impulse responses, or within-period timing.

1. Define the target and the native frequency of each information series.
2. Map publication lags and real-time availability.
3. Decide whether the goal is forecasting/nowcasting or structural analysis.
4. Choose MIDAS, stacked MF-VAR, or a state-space representation.
5. Preserve within-period timing whenever it is economically meaningful.
6. Evaluate forecasts in pseudo-real time and structural results against aggregation choices.
7. Check robustness to lag weights, release timing and alternative identification.

1. Mixed-frequency data are the norm, not the exception, in macroeconomics.
2. Temporal aggregation can create information loss and structural bias.
3. MIDAS offers a parsimonious way to exploit many high-frequency lags.
4. MF-VARs preserve multivariate feedback and within-period timing for structural analysis.
5. Nowcasting requires models that respect ragged edges, publication lags and sequential data releases.

- ▶ Compare Beta, exponential Almon and U-MIDAS weights using another monthly predictor.
- ▶ Re-estimate the MF-VAR with industrial production or inflation added to the system.
- ▶ Compare month-1 and month-3 shocks and discuss the economic interpretation of timing.
- ▶ Build a pseudo-real-time GDP nowcast and update it as monthly releases arrive.
- ▶ Compare MF-VAR IRFs with a quarterly VAR based on averaged monthly data.
- ▶ Discuss how data revisions change forecast evaluation using first-release versus final-vintage GDP.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 11. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your own mixed-frequency datasets and release calendars.
- ▶ Add Python, R or MATLAB demonstrations.
- ▶ Add real-time vintages, nowcasting dashboards and news decompositions.
- ▶ Add alternative identification schemes and structural applications.

Make them yours.

Quantile Methods and Growth-at-Risk

Moving from the conditional mean to the full distribution

Next question

How can macroeconomic models quantify downside risks and tail outcomes rather than focusing only on average responses?